National University of Computer and Emerging Sciences

Electro Mechanical Systems (EE2010)

Date: May 25th, 2024

Course Instructor(s)

Mr. Muhammad Abdul Majid

Mr. Syed Muhammad Ismaeel

Final Exam

Total Time: 3 Hours
Total Marks: 100
Total Questions: 4

Semester: SP-2024
Campus: Lahore
Dept: Electrical Engineering

Student Name	Roll No	Section	Student Signature
Arshaq Waseem.	22L-7697	4C	

Vetted by _____ Vetter Signature _____

CLO #3: Analyze Synchronous Generator performance along with special emphasis towards environmental cost of generation

Q1: **[marks 25]**

A **2.8 KV, 1.5 MVA,** 0.85-PF-lagging, 50-Hz, 16 pole, Y-connected synchronous generator, has a synchronous reactance of 1.2 Ω and an armature resistance of 0.2 Ω. At 50 Hz, its friction, windage and core losses are 50 KW. The field circuit has a dc voltage of 200 V. The resistance of the field circuit is adjustable over the range from 10 Ω to 210 Ω.

The generator is providing 300A, at 0.8 PF lagging to a load. The line voltage at load terminals is 2500 V. Answer the following questions.

1443.4 V

(a) Draw a properly labelled equivalent circuit diagram with proper current directions and voltage polarities
(b) Find Voltage regulation
(c) Field circuit resistance
(d) Output torque of prime mover

I_F (A)	2	4	6	8	10
$V_{T,NL}$ (V)	1200	2200	2700	2950	3050

OCC for above generator

2500.

$I_F : V_T$
$2 : 250$
$x : 100$

$x = \frac{200}{250}$
0.8
$I_F = 6.8A$

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CLO #2: Investigate working of a DC Machine

Q2: **[marks =25]**

A 25HP, 240V compensated DC shunt motor has armature resistance of 0.4 Ω. Its field resistance is 80 Ω and adjustable resistance connected in series with field coil can vary from 50 Ω to 200 Ω. Its no load characteristic curve at a speed of 1800 rpm is tabulated as under.

I_F (A)	0.3	0.6	0.9	1.2	1.4	1.6
E_A (V)	130	230	274	288	292	295

Find out following quantities assuming rated terminal voltage in all cases.

a) No load speed of motor if adjustable resistance connected in series with field coil has been set to a value of 100 ohm.
b) The speed at a line current of 65A, with same field current as in part (a)
c) Fully labeled Circuit diagram for part (b)
d) Output torque in part (b) if mechanical and core losses are 700 watts
e) Maximum possible no load speed of the motor

CLO #5: Analyze an Induction motor circuit

Q3: **[marks =25]**

A 120 HP, 440-V, 50-Hz four-pole Y-connected induction motor is working under steady conditions. The equivalent circuit parameters are

$R_1 = 0.075\ \Omega$ $R_2 = 0.065\ \Omega$ $X_M = 7.2\ \Omega$
$X_1 = 0.17\ \Omega$ $X_2 = 0.17\ \Omega$
$P_{F\&W} = 1.0\ kW$ $P_{misc} = 150\ W$ $P_{CORE} = 1.1\ Kw$

For a slip of 5% and at rated voltage and frequency, find out

a) Motor speed
b) The stator copper losses
c) The air gap power
d) Shaft torque

CLO #4: Appraise the existence of a rotating magnetic field and its application to a Synchronous Motor

Q4: **[marks =25]**

A 415-V, Y-connected, 30 pole, synchronous motor is delivering a power of 30 HP at 0.75 lagging power factor and rated voltage. The field current flowing under these conditions is 4.5 A. Its synchronous reactance is 1.2 Ω. Total losses of the motor under these conditions are 2300watts. Assume a linear open-circuit characteristic.

Find the torque angle and internally generated voltage

How much field current is required to run the motor at a PF of 0.8 leading, keeping everything else constant?

What is the load torque?

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